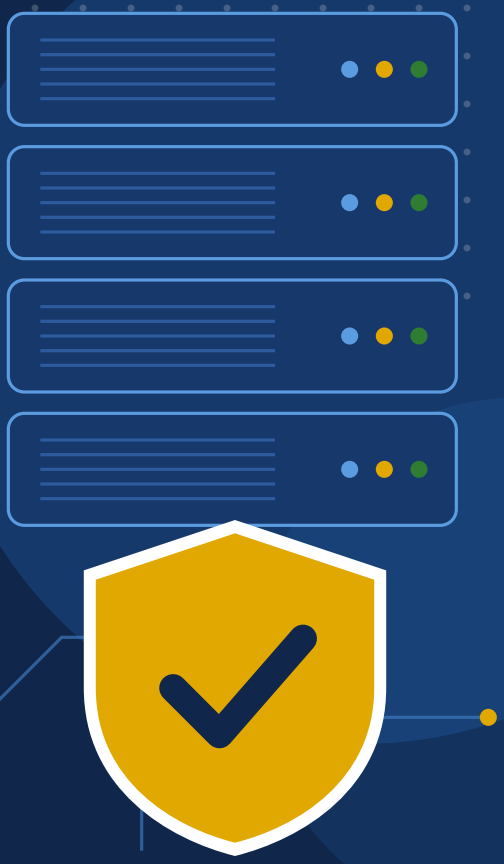
SCHOOL

UNIVERSITY OF PANGASINAN

IT Services Department



UNIVERSITY ENROLLMENT SYSTEM

Production Outage Recovery Runbook

A step-by-step guide for restoring enrollment services quickly and safely, even when the senior administrator is unavailable.

SEV-1 | CRITICAL INCIDENT RESPONSE

DOCUMENT ID	SYSTEM	ENVIRONMENT	VERSION
IT-RUNBOOK-001	UPang Online Enrollment System (UPang-OES)	Production	1.0

OWNER	LAST REVIEWED	NEXT REVIEW	CLASSIFICATION
UPang IT Services Department	October 3, 2026	January 3, 2027	Internal Use

SCHOOL

University of Pangasinan

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Junior IT Engineer

Document Control

Document ID	IT-RUNBOOK-001	**Version**	1.0
System	UPang Online Enrollment System (UPang-OES)	**Environment**	Production
Owner	UPang IT Services Department	**Classification**	Internal Use
Last Reviewed	October 3, 2026	**Next Review**	January 3, 2027

About the sample values. Every bracketed placeholder in the original template has been filled in with illustrative University of Pangasinan data. **Values shown in blue** are the filled-in entries. Replace them with real production details before use.

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1 Purpose

This runbook explains what to do when the university's enrollment system becomes unavailable.

It is designed for situations where:

- The enrollment system is offline.
- The normal system administrator is unavailable.
- A junior IT engineer is responding.
- The cause of the outage is initially unknown.
- Recovery must happen quickly and safely.

PRIMARY GOAL

Restore the enrollment system to a known-good state while protecting student and university data.

SECONDARY GOAL

Record what happened so the next engineer does not have to start from zero.

2 Incident Definition

An incident is an unexpected problem that requires urgent action. incident.io recommends declaring an incident when one or more of these are present:

- Negative impact
- Urgency
- Coordination between people
- Stakeholder communication

This situation is a **SEV-1 / Critical Incident** if:

- Enrollment is completely unavailable.
- A large number of students cannot enroll.
- University operations are significantly affected.
- There is no immediate workaround.

THIS INCIDENT

SEV-1
CRITICAL
Enrollment completely unavailable; no workaround.

SEV-2
MAJOR
Partial outage or serious slowdown; limited workaround.

SEV-3
MINOR
Small issue with limited user impact.

Incident Severity: SEV-1 (Critical)

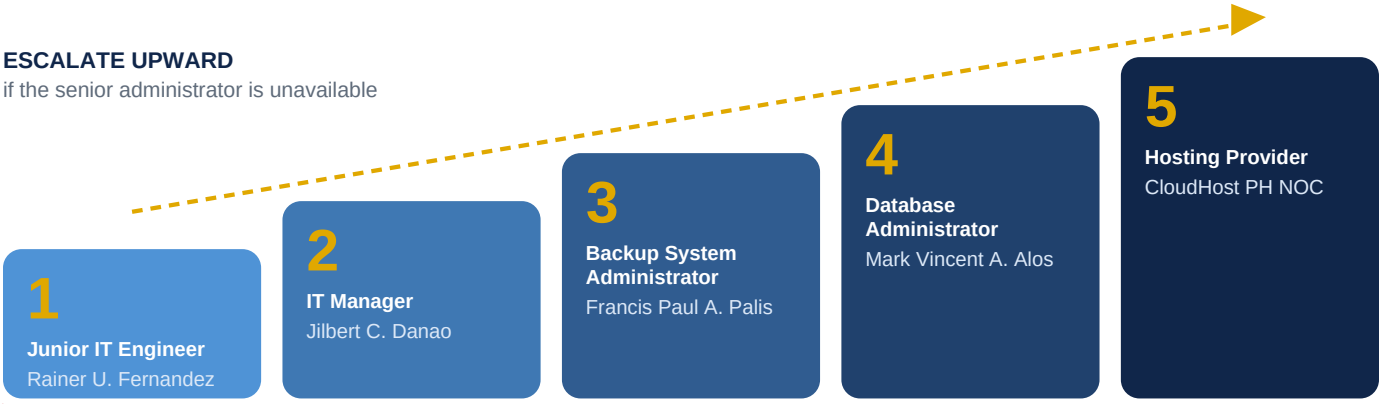
3 Important Contacts

Never put actual passwords, API keys, or private credentials in this document.

Role	Name	Contact
Junior IT Engineer	Rainer U. Fernandez	0917-555-0101
Senior System Administrator	Kenley Russel S. Benitez	0917-555-0102
IT Manager	Jilbert C. Danao	0917-555-0103
Database Administrator	Mark Vincent A. Alos	0917-555-0104
Network Administrator	Francis Paul A. Palis	0917-555-0105
University Registrar	Ainslee A. Culhi	0917-555-0106
Enrollment Office	Office of the Registrar - Enrollment Desk	(075) 555-0120 / enrollment@upang.edu.ph
Security Officer	Sir Xziann Jeano P. Genuino	0917-555-0107
Server Hosting Provider	CloudHost PH (sample provider)	24/7 NOC: (02) 8555-0140
Internet/ISP Provider	NetLink Broadband (sample provider)	Enterprise support: (02) 8555-0150

Emergency escalation

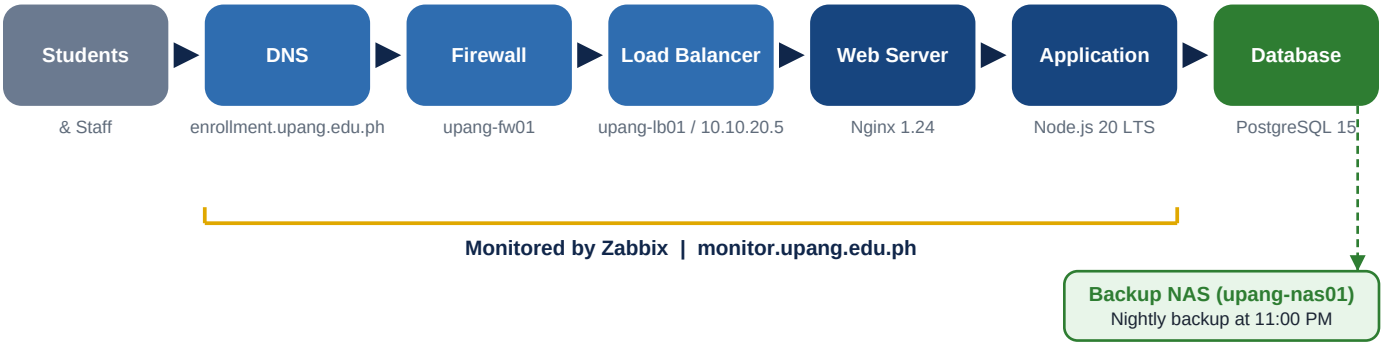
If the senior administrator is unavailable:



Do not wait indefinitely for one person to respond.

4 System Information

Complete this section before an incident occurs.



Component	Information
Application	UPang Online Enrollment System (UPang-OES)
Production URL	https://enrollment.upang.edu.ph
Application Server	upang-app01 / 10.10.20.11
Database Server	upang-db01 / 10.10.20.21
Database	PostgreSQL 15
Operating System	Ubuntu Server 22.04 LTS
Web Server	Nginx 1.24
Application Runtime	Node.js 20 LTS
Load Balancer	upang-lb01 / 10.10.20.5
Firewall	upang-fw01 (pfSense edge firewall)
Monitoring	Zabbix - https://monitor.upang.edu.ph
Backup System	Veeam Backup, nightly at 11:00 PM
Latest Known Good Backup	upang-nas01:/backups/oes/daily/ (Oct 2, 2026 11:00 PM)
Source Repository	https://git.upang.edu.ph/itsd/oes
Deployment System	GitLab CI/CD pipeline
Status Page	https://status.upang.edu.ph
Incident Channel	Microsoft Teams: UPang IT Incident Response

5 Before an Incident Happens

The recovery procedure is only useful if the information needed to perform it actually exists. The IT department should maintain:

- [ ] Current server inventory
- [ ] Current network diagram
- [ ] Database backup schedule
- [ ] Backup restoration instructions
- [ ] Application deployment procedure
- [ ] Server access procedure
- [ ] Monitoring information
- [ ] Contact list
- [ ] Escalation list
- [ ] Known-good application version
- [ ] Known-good database backup
- [ ] Disaster recovery procedure
- [ ] Maintenance procedures

> The most important requirement is that a junior engineer can find this information **without asking the unavailable senior administrator**.

6 2:00 AM Incident — First 5 Minutes

Step 1 — Confirm the outage

> **Do not immediately restart the server.** First determine whether the system is actually unavailable.

Check:

- [ ] Enrollment website
- [ ] Application login
- [ ] DNS
- [ ] Network connectivity
- [ ] Server availability
- [ ] Monitoring dashboard
- [ ] Recent alerts

Record the results.

Incident log

```
[02:00] Enrollment system reported unavailable.
[02:02] Confirmed application URL is unreachable.
[02:03] Monitoring dashboard checked.
[02:04] Server connectivity test performed.
```

7 Declare the Incident

Create an incident using the university's incident-management process.

Incident	**[UPang-OES] Production Outage**
Severity	**SEV-1**
Started	**October 3, 2026 - 02:00 AM (PHT)**
Detected by	**Rainer U. Fernandez**

Current impact: Students and staff are currently unable to access the university enrollment system.

incident.io recommends creating a dedicated shared space early in the response so responders have one place for coordination and information.

8 Create the Incident Channel

Create:

Channel name	**#incident-enrollment-2026-10-03**
or Teams channel	**UPang IT Incident Response**

Use this channel as the central source of truth. Post:

- Current symptoms
- People involved
- Actions being performed
- Findings
- Decisions
- Recovery status
- Important timestamps

Avoid splitting the investigation across many private conversations. The incident channel becomes the shared memory of the incident.

9 Assign Roles

Do not let everyone investigate the same thing. For a small university IT team:

INCIDENT LEAD	TECHNICAL LEAD	COMMUNICATIONS	SCRIBE
JD	RF	AC	SG
Jilbert C. Danao	**Rainer U. Fernandez**	**Ainslee A. Culhi**	**Sir Xziann Jeano P. Genuino**
IT Manager	*Junior IT Engineer*	*University Registrar*	*Security Officer*
Coordinates the incident	Investigates and performs technical recovery	Updates registrar, management and users	Records timeline and actions

If only one engineer is available: Engineer = Technical Lead. The engineer should still identify someone to handle communication if possible. The incident lead should coordinate rather than becoming distracted by technical troubleshooting.

10 Initial Communication

Send a short message to the appropriate stakeholders.

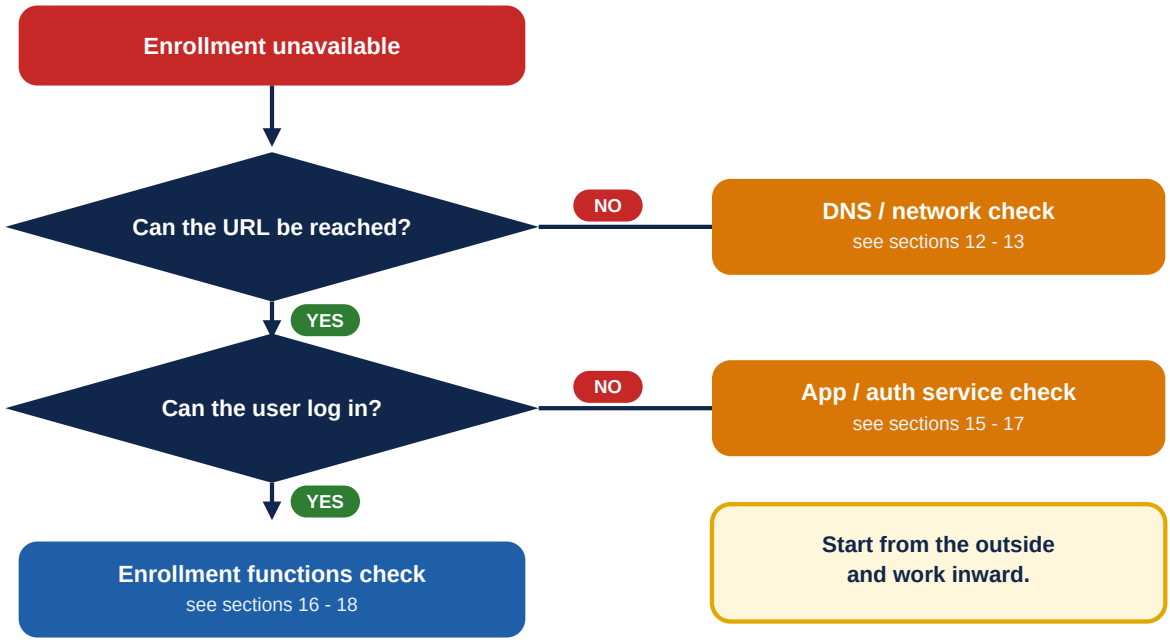
INCIDENT: Enrollment System Unavailable

The university enrollment system is currently unavailable.
Detected: 02:00 AM
Impact: Students and staff cannot currently access enrollment services.
Status: Investigating
Incident Lead: Jilbert C. Danao
Technical Lead: Rainer U. Fernandez
The IT team is investigating the cause and working to restore service.
Next update: 02:30 AM

- Do not guess the cause.
- Do not say “the server crashed” until this has been verified.

11 Troubleshooting Decision Tree

Start from the outside and work inward.



12 Check 1 — DNS

Verify that **enrollment.upang.edu.ph** resolves to the expected address. Check:

- [ ] DNS record
- [ ] Domain expiration
- [ ] DNS provider status
- [ ] Recent DNS changes

If DNS is incorrect: do **not** randomly modify DNS records. Contact the Network Administrator, **Francis Paul A. Palis** (**0917-555-0105**).

Record:

```
Expected:      203.0.113.25
Actual:        203.0.113.25   (MATCH)

Time checked:  02:04 AM
```

13 Check 2 — Network

Determine whether the server can be reached. Check:

- [ ] Network connection
- [ ] Router
- [ ] Load balancer
- [ ] Firewall
- [ ] VPN
- [ ] Server connectivity

If other university systems are also unavailable, suspect a wider infrastructure / network incident.

If only enrollment is unavailable, continue with the application / server investigation.

Result in this scenario: **other university systems are normal, so the investigation continues at the server.**

14 Check 3 — Server

Verify:

- [ ] Server is powered on
- [ ] Server is reachable
- [ ] Disk is not full
- [ ] Memory is not exhausted
- [ ] CPU is not permanently overloaded
- [ ] Important services are running

Record observations before making changes.

SERVER: upang-app01 (10.10.20.11)

CPU		**18%**	Reachable	**YES**
Memory		**41%**	Application service	**STOPPED**
Disk		**63%**	Database service	**RUNNING**

15 Check 4 — Application Service

Determine whether the application process is running.

APPLICATION SERVICE	WEB SERVER	APPLICATION RUNTIME	DATABASE
FAILED	RUNNING	STOPPED	RUNNING

> Use the university's documented service-management procedure. **Do not invent service names or commands.**

If the application service has stopped unexpectedly:

1	Check logs.
2	Record the error.
3	Confirm the database is available.
4	Check whether the server has enough resources.
5	Restart the application service if the restart is documented and low-risk.
6	Verify the application.

16 Check 5 — Database

The database is one of the most important components. Check:

- [ ] Database service status
- [ ] Database connectivity
- [ ] Disk space
- [ ] Database errors
- [ ] Connection limits
- [ ] Recent database changes
- [ ] Backup status

> **Important rule: do not restore a database backup just because the application is offline.** First determine whether the existing database is healthy. Restoring a backup unnecessarily can cause data loss.

17 Check 6 — Application Logs

Check the application logs around the time the outage began. Look for:

- [ ] Database connection errors
- [ ] Authentication failures
- [ ] Configuration errors
- [ ] Permission errors
- [ ] Disk errors
- [ ] Memory errors
- [ ] Failed deployments
- [ ] Certificate errors
- [ ] Dependency failures

Record the relevant error **without copying sensitive information** into the incident channel. Example:

```
02:17 - Application log shows repeated database connection failures.

02:19 - Database server is reachable.

02:20 - Database service is RUNNING, but rejects the application login
        (invalid credentials in the new application configuration).
```

18 Check 7 — Recent Changes

Ask: What changed shortly before the outage?

Check:

- [ ] Application deployment
- [ ] Server updates
- [ ] Database changes
- [ ] Firewall changes
- [ ] DNS changes
- [ ] Certificate renewal
- [ ] Configuration changes
- [ ] Network changes

If the outage started immediately after a known deployment, consider rolling back to the **known-good version**. incident.io recommends considering rollback to a known-good version rather than spending incident time creating a new fix under pressure.

Finding in this scenario: release **v2.5.0** was deployed at **01:45 AM**; the outage began about two minutes later. Known-good rollback target: **v2.4.1**.

19 Recovery Priority

Use this order:

1 — Restore service
Get enrollment working again.

2 — Protect data
Do not perform destructive operations unless necessary.

3 — Investigate root cause
Understand why the outage occurred.

4 — Permanent fix
Handle longer-term improvements after the immediate incident.

The immediate goal during an incident is mitigation; larger cleanup or engineering work can become follow-up tasks after service is restored.

20 Common Recovery Actions

Only perform an action if it is **documented** and appropriate for the observed failure.

Problem	Possible Action
Application stopped	Restart application service
Web server stopped	Restart web server
Database stopped	Restart database service after checking logs
Disk full	Remove approved temporary files / expand storage
Bad deployment	Roll back to known-good version
Configuration error	Restore known-good configuration
Expired certificate	Follow certificate replacement procedure
Network problem	Escalate to network administrator
Hardware failure	Fail over to secondary server
Database corruption	Follow database recovery procedure
Server failure	Fail over / restore server
Unknown cause	**Escalate**

21 Do Not Do These Things

A junior engineer should **not**:

- Delete database files.
- Delete production data.
- Run untested database commands.
- Modify firewall rules randomly.
- Disable security controls just to make the system work.
- Restore an old backup without checking its age.
- Change multiple unrelated systems simultaneously.
- Deploy experimental code directly to production.
- Guess passwords or credentials.
- Share credentials in chat.
- Hide failed recovery attempts.
- Claim the incident is resolved without testing.

If an action could cause irreversible data loss, stop and escalate.

22 Database Recovery

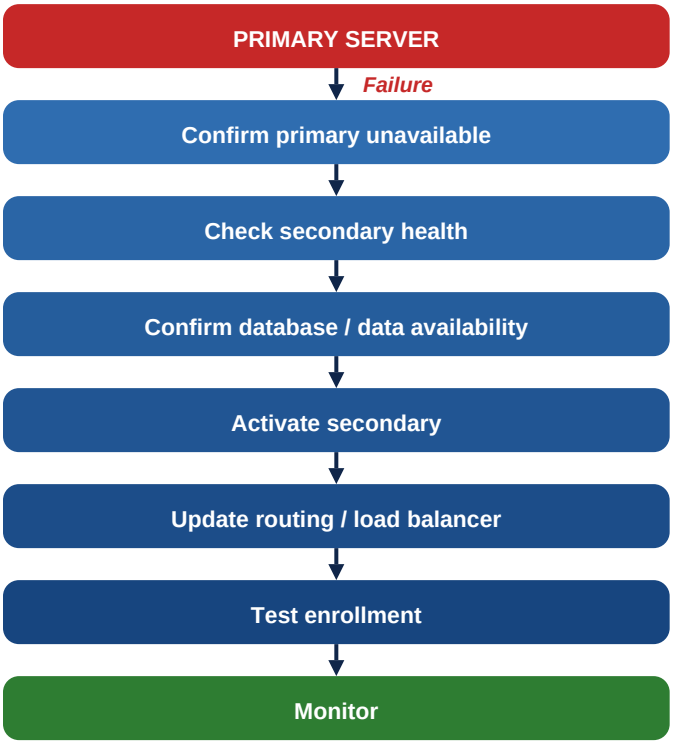
Database recovery should only happen when the database itself is confirmed to be damaged or unavailable and normal recovery is not possible.

Before restoring a backup

Confirm before restoring	Answer	Detail
Backup exists	YES	Veeam nightly backup
Backup date	Oct 2, 2026	
Backup time	11:00 PM	
Backup verified	YES	Restore test passed
Database failure confirmed	NO	Database is healthy
Data-loss risk understood	YES	Up to 3 hours of enrollments
Approval obtained	N/A	Restore not required

If any critical information is unknown: **escalate to the Database Administrator (Mark Vincent A. Alos) or the IT Manager (Jilbert C. Danao).**

23 Failover Procedure



If a secondary server exists, follow this sequence from top to bottom. Do not skip a step.

Never assume that the secondary server contains current data. Verify synchronization first.

24 Recovery Verification

Do not declare the system recovered simply because the server responds. **Perform an end-to-end test.**

Technical checks

- Website loads
- Login works
- Student account can authenticate
- Enrollment page loads
- Course search works
- Enrollment transaction works in test mode
- Database writes work
- Confirmation / receipt works
- Logs show no continuing critical errors
- Monitoring is healthy

User-level test

Ask an authorized test user to perform the normal enrollment workflow:



25 Monitor After Recovery

Do not immediately close the incident. Monitor for **45 minutes** (recommended range: 30–60 minutes). Check:

- CPU
- Memory
- Disk
- Database
- Application errors
- Login failures
- Transaction failures
- Network
- User reports

The system should remain stable before declaring full recovery.

26 Recovery Communication

RESOLVED: Enrollment System

The university enrollment system has been restored.
Service restored: 02:35 AM
The IT team has completed initial verification of: system accessibility, user login, enrollment functions and database connectivity.
The system is currently being monitored for stability.
A follow-up review will be conducted to determine the cause of the outage and identify preventive actions.

27 Close the Incident

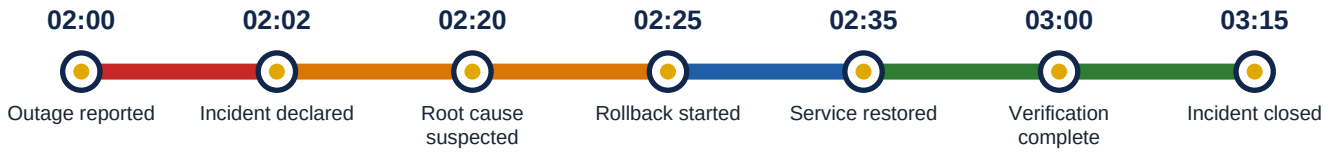
The incident can be closed when:

- Enrollment is operational.
- Critical functionality has been tested.
- No active critical errors remain.
- Monitoring is normal.
- Stakeholders have been notified.
- Incident timeline is complete.
- Remaining work has been assigned.

Closing marks the point where immediate impact is mitigated; longer-term follow-up work moves into normal team workflows rather than leaving the incident open indefinitely.

28 Incident Timeline

The scribe should maintain this throughout the incident. Sample timeline from the 2:00 AM scenario:



DETECT | DIAGNOSE | RECOVER | VERIFY & MONITOR

Time	Event	Person	Action / Result
02:00	Outage reported	**Rainer U. Fernandez**	Enrollment unavailable
02:02	Incident declared	**Rainer U. Fernandez**	SEV-1
02:05	Server checked	**Rainer U. Fernandez**	**upang-app01 reachable; application service FAILED**
02:10	Database checked	**Mark Vincent A. Alos**	**Database healthy; no restore needed**
02:15	Logs checked	**Rainer U. Fernandez**	**Repeated database login failures after 01:45 deployment**
02:20	Root cause suspected	**Rainer U. Fernandez**	**Invalid credentials in release v2.5.0 configuration**
02:25	Recovery started	**Rainer U. Fernandez**	**Rollback to v2.4.1 approved by IT Manager**
02:35	Service restored	**Rainer U. Fernandez**	**Application running on v2.4.1**
03:00	Verification complete	**Rainer U. Fernandez**	**Test student completed an enrollment**
03:15	Incident closed	**Jilbert C. Danao**	Monitoring normal

29 Post-Incident Review

After the system is stable, review:

What happened?	**Release v2.5.0 was deployed at 01:45 AM with an invalid database credential in its configuration. The application could not reach the database and enrollment became unavailable.**
When did it start?	**October 3, 2026, about 01:47 AM**
How was it detected?	**Monitoring alert (Zabbix) at 02:00 AM**
What was affected?	**UPang-OES web portal, student login and enrollment transactions. The database and its data were not affected.**
What caused it?	**Configuration error in release v2.5.0**
How was it recovered?	**Rolled back to known-good version v2.4.1, restored the known-good configuration and restarted the application service.**
What went well?	**Fast detection, quick SEV-1 declaration, clear roles, and no unnecessary database restore.**
What went poorly?	**The configuration was not validated before the production deployment.**
What was missing?	**Pre-deployment configuration check, automated rollback and a documented backup-administrator procedure.**

Follow-up actions

Action	Owner	Priority	Due Date
Create missing recovery documentation	**Kenley Russel S. Benitez**	**High**	**Oct 16, 2026**
Improve monitoring	**Francis Paul A. Palis**	**High**	**Oct 23, 2026**
Test backup restoration	**Mark Vincent A. Alos**	**High**	**Oct 30, 2026**
Document server architecture	**Rainer U. Fernandez**	**Medium**	**Nov 13, 2026**
Train backup administrator	**Jilbert C. Danao**	**High**	**Nov 20, 2026**

> The objective should be **learning and improving the system** rather than blaming the person who happened to be responding.

ENROLLMENT SYSTEM DOWN

The “Junior Engineer at 2 AM” Quick Card | print this page and keep it separately

TELL STAKEHOLDERS: Investigating ▶ Recovering ▶ Monitoring ▶ Resolved

1 Confirm
Is it actually down? Check the website, network, monitoring and server. Do not restart anything yet.

2 Declare
Create the incident (SEV-1). Channel: #incident-enrollment-2026-10-03 | Teams: UPang IT Incident Response.

3 Assign
Incident Lead: Jilbert C. Danao. Technical Lead: Rainer U. Fernandez. Comms: Ainslee A. Culhi. Scribe: Sir Xziann Jeano P. Genuino.

4 Check
Work from the outside in, in the order shown in the chain below. Record every observation.

5 Recover
Use the lowest-risk documented recovery action. Known-good rollback beats an experimental fix.

6 Verify
Test the complete enrollment process end to end with an authorized test user.

7 Monitor
Watch CPU, memory, disk, database, errors and logins for 45 minutes after recovery.

8 Communicate
Investigating, Recovering, Monitoring, Resolved. Always give the time of the next update.

9 Record
What happened, what was checked, what was changed, what fixed it, and when.

10 Close
Only close after verification, normal monitoring and stakeholder notification.

STEP 4: CHECK IN THIS ORDER

DNS ▶ Network ▶ Server ▶ Web server ▶ Application ▶ Database ▶ Logs ▶ Recent changes

EMERGENCY CONTACTS

IT MANAGER	SENIOR SYSADMIN	DATABASE ADMIN	INCIDENT CHANNEL
Jilbert C. Danao	**Kenley Russel S. Benitez**	**Mark Vincent A. Alos**	**#incident-enrollment-2026-10-03**
0917-555-0103	0917-555-0102	0917-555-0104	Teams: UPang IT Incident Response

CRITICAL RULE

If you don't know what a command or recovery action will do, don't run it on production.

Stop, document what you know, and escalate to the IT Manager (Jilbert C. Danao, 0917-555-0103).

31 Critical Rule



If you don't know what a command or recovery action will do, don't run it on production.

Stop, document what you know, and escalate to: **IT Manager - Jilbert C. Danao (0917-555-0103)**, the Senior System Administrator (**Kenley Russel S. Benitez**), the DBA (**Mark Vincent A. Alos**) or another approved escalation contact.

A good recovery runbook should make the safe path obvious even to someone who did not build the system. That is particularly important here because the scenario specifically assumes that the person who normally knows the server is unavailable.

Design principle: the documentation should be written so that a junior engineer can follow it at 2:00 AM without relying on someone's memory. That means consistent incident declaration, clear ownership, a shared incident space, explicit roles, verified evidence, mitigation first, and a documented timeline.

Sources

- incident.io — Defining an incident (incident.io/guide/foundations/defining-an-incident)
- incident.io — The lifecycle of an incident (incident.io/guide/response/the-lifecycle-of-an-incident)
- incident.io — Working with distributed teams (incident.io/guide/response/working-with-distributed-teams)